

MODULE 05 — ASSIGNMENT

Problem Solving with AI (Claude Code + JavaScript)

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Course: AI Driven SQA — Manual & Automation Testing | OSTAD

Q1. JavaScript Coding Practice

The five programs below solve common problem-solving exercises in JavaScript. Each includes a problem statement, source code, sample input, expected output, a real execution screenshot, and a short explanation.

1. Convert Celsius to Fahrenheit**PROBLEM STATEMENT**

Write a function that takes a temperature in Celsius and converts it to Fahrenheit using the formula: $F = (C \times 9/5) + 32$.

SOURCE CODE

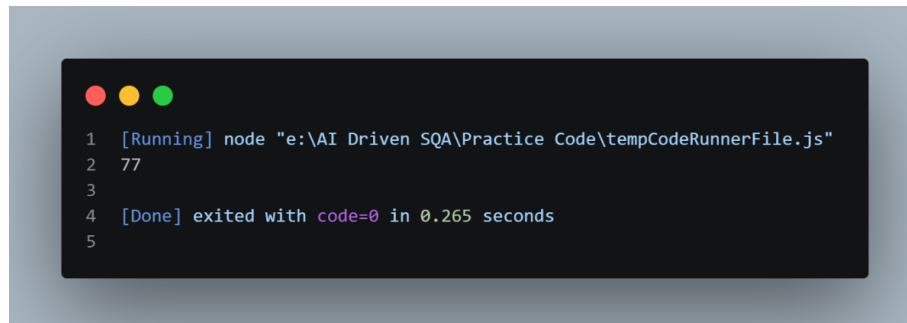
```
function celsiusToFahrenheit(celsius) {  
  return (celsius * 9 / 5) + 32;  
}  
  
console.log(celsiusToFahrenheit(25));
```

INPUT

celsiusToFahrenheit(25)

EXPECTED OUTPUT

77

OUTPUT

HANDS-ON PRACTICE

Executed live in VS Code to confirm the result independently.

```

File Edit Selection View Go Run ... Practice Code
Explorer
  Pract...
  > Code Practise
  JS 1stclass.js
  JS 5thModuleAssignme...
  JS Assignment Module ...
  JS logintest.js
  JS LoginTestFeature.js

  > Outline
  > Timeline 5thModuleAss...
  Local History will track recent changes as you save them unless the file has been excluded or is too large. Source

JS 5thModuleAssignment.js > ...
1 function celsiusToFahrenheit(celsius) {
2   return (celsius * 9 / 5) + 32;
3 }
4
5 console.log(celsiusToFahrenheit(25));
6

Problems Output Debug Console Terminal Ports
[Running] node "e:\AI Driven SQA\Practice Code\tempCodeRunnerFile.js"
77
[Done] exited with code=0 in 0.265 seconds

```

EXPLANATION

The formula multiplies the Celsius value by 9/5 (which scales the temperature to the Fahrenheit degree size) and then adds 32 (since the two scales don't share a zero point). For 25°C: $25 \times 9/5 = 45$, then $45 + 32 = 77^\circ\text{F}$.

2. Find the Factorial of a Number

PROBLEM STATEMENT

Write a function that calculates the factorial of a given non-negative integer (e.g., $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$).

SOURCE CODE

```

function factorial(n) {
  if (n === 0 || n === 1) return 1;
  let result = 1;
  for (let i = 2; i <= n; i++) {
    result *= i;
  }
  return result;
}

console.log(factorial(5));

```

INPUT

factorial(5)

EXPECTED OUTPUT

120

OUTPUT

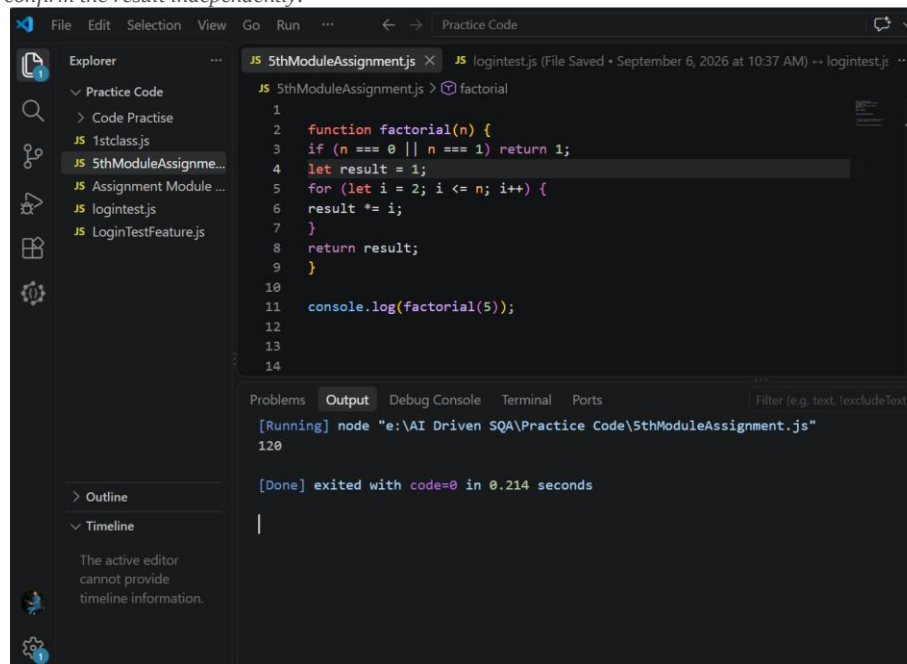
```

1 [Running] node "e:\AI Driven SQA\Practice Code\5thModuleAssignment.js"
2 120
3
4 [Done] exited with code=0 in 0.214 seconds
5

```

HANDS-ON PRACTICE

Executed live in VS Code to confirm the result independently.



EXPLANATION

0! and 1! are defined as 1, so those are handled as a base case. For any larger n, the loop multiplies result by every integer from 2 up to n. For 5: $1 \times 2 \times 3 \times 4 \times 5 = 120$

3. Check for Palindrome

PROBLEM STATEMENT

Write a function that checks if a given word (e.g., "racecar") reads the same backward as forward and returns true or false.

SOURCE CODE

```

function isPalindrome(word) {
  const cleaned = word.toLowerCase();
  const reversed = cleaned.split("").reverse().join("");
  return cleaned === reversed;
}

console.log(isPalindrome("racecar"));

```

INPUT

isPalindrome("racecar")

EXPECTED OUTPUT

true

OUTPUT

```

1  [Running] node "e:\AI Driven SQA\Practice Code\5thModuleAssignment.js"
2  true
3
4  [Done] exited with code=0 in 0.238 seconds

```

HANDS-ON PRACTICE

Executed live in VS Code to confirm the result independently.

The screenshot shows the VS Code interface with the file `5thModuleAssignment.js` open. The code defines a function `isPalindrome(word)` that checks if a word is a palindrome by converting it to lowercase, splitting it into characters, reversing the order, and joining it back. It then logs the result of `isPalindrome("racecar")`. The Output panel at the bottom shows the execution result: `[Running] node "e:\AI Driven SQA\Practice Code\5thModuleAssignment.js"` followed by `true` and `[Done] exited with code=0 in 0.238 seconds`.

EXPLANATION

Converting to lowercase first makes the check case-insensitive. `split("")` turns the string into an array of characters, `.reverse()` flips the order, and `join("")` puts it back together as a string. If the reversed version matches the original exactly, the word is a palindrome — "racecar" reversed is still "racecar", so the function returns true.

4. Sum of Array Elements

PROBLEM STATEMENT

Write a function that takes an array of numbers and returns the sum of all elements in the array.

SOURCE CODE

```

function sumArray(arr) {
  return arr.reduce((total, num) => total + num, 0);
}

console.log(sumArray([10, 20, 30, 40]));

```

INPUT

`sumArray([10, 20, 30, 40])`

EXPECTED OUTPUT

100

OUTPUT

```

1  [Running] node "e:\AI Driven SQA\Practice Code\tempCodeRunnerFile.js"
2  100
3
4  [Done] exited with code=0 in 0.212 seconds
5

```

HANDS-ON PRACTICE

Executed live in VS Code to confirm the result independently.

```

JS 5thModuleAssignment.js
1  function sumArray(arr) {
2  return arr.reduce((total, num) => total + num, 0);
3  }
4
5  console.log(sumArray([10, 20, 30, 40]));
6
7
8
9
10
11

```

```

[Running] node "e:\AI Driven SQA\Practice Code\tempCodeRunnerFile.js"
100

[Done] exited with code=0 in 0.212 seconds

```

EXPLANATION

reduce() walks through the array once, carrying a running total (starting at 0) and adding each element to it in turn. For [10, 20, 30, 40], the total accumulates as $0 \rightarrow 10 \rightarrow 30 \rightarrow 60 \rightarrow 100$.

5. FizzBuzz (Basic)

PROBLEM STATEMENT

Write a loop from 1 to 15 that prints "Fizz" for numbers divisible by 3, "Buzz" for numbers divisible by 5, "FizzBuzz" for numbers divisible by both, and the number itself otherwise.

SOURCE CODE

```

for (let i = 1; i <= 15; i++) {
  if (i % 3 === 0 && i % 5 === 0) {
    console.log("FizzBuzz");
  } else if (i % 3 === 0) {
    console.log("Fizz");
  } else if (i % 5 === 0) {
    console.log("Buzz");
  } else {
    console.log(i);
  }
}

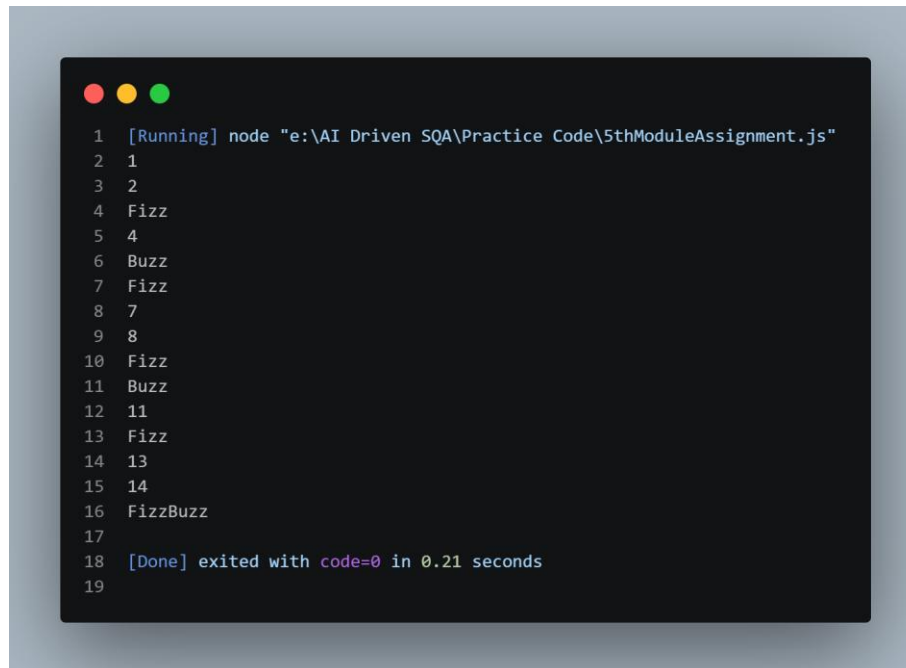
```

INPUT

Loop from 1 to 15 (no external input)

EXPECTED OUTPUT

1, 2, Fizz, 4, Buzz, Fizz, 7, 8, Fizz, Buzz, 11, Fizz, 13, 14, FizzBuzz

OUTPUT


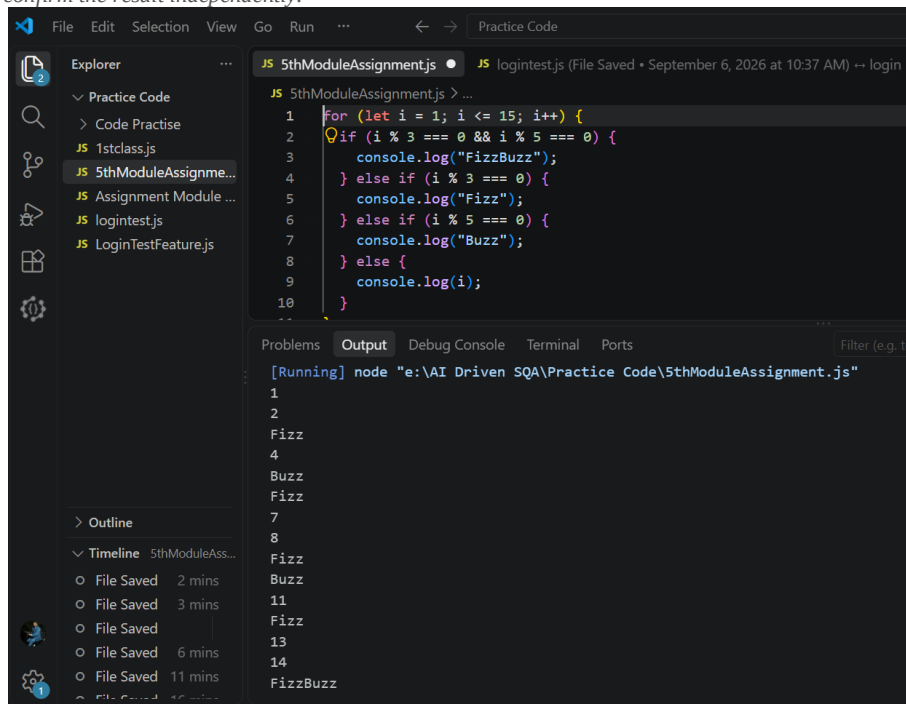
```

1 [Running] node "e:\AI Driven SQA\Practice Code\5thModuleAssignment.js"
2 1
3 2
4 Fizz
5 4
6 Buzz
7 Fizz
8 7
9 8
10 Fizz
11 Buzz
12 11
13 Fizz
14 13
15 14
16 FizzBuzz
17
18 [Done] exited with code=0 in 0.21 seconds
19

```

HANDS-ON PRACTICE

Executed live in VS Code to confirm the result independently.



```

JS 5thModuleAssignment.js > ...
1 for (let i = 1; i <= 15; i++) {
2   if (i % 3 === 0 && i % 5 === 0) {
3     console.log("FizzBuzz");
4   } else if (i % 3 === 0) {
5     console.log("Fizz");
6   } else if (i % 5 === 0) {
7     console.log("Buzz");
8   } else {
9     console.log(i);
10  }
}

```

Problems Output Debug Console Terminal Ports Filter (e.g. test)

```

[Running] node "e:\AI Driven SQA\Practice Code\5thModuleAssignment.js"
1
2
Fizz
4
Buzz
Fizz
7
8
Fizz
Buzz
11
Fizz
13
14
FizzBuzz

```

EXPLANATION

The check for both 3 and 5 must come first, otherwise a multiple of 15 would only ever match the first true condition (3 or 5) and never reach "FizzBuzz". Each number from 1 to 15 is tested in order: multiples of 15 print "FizzBuzz", other multiples of 3 print "Fizz", other multiples of 5 print "Buzz", and everything else prints the plain number.